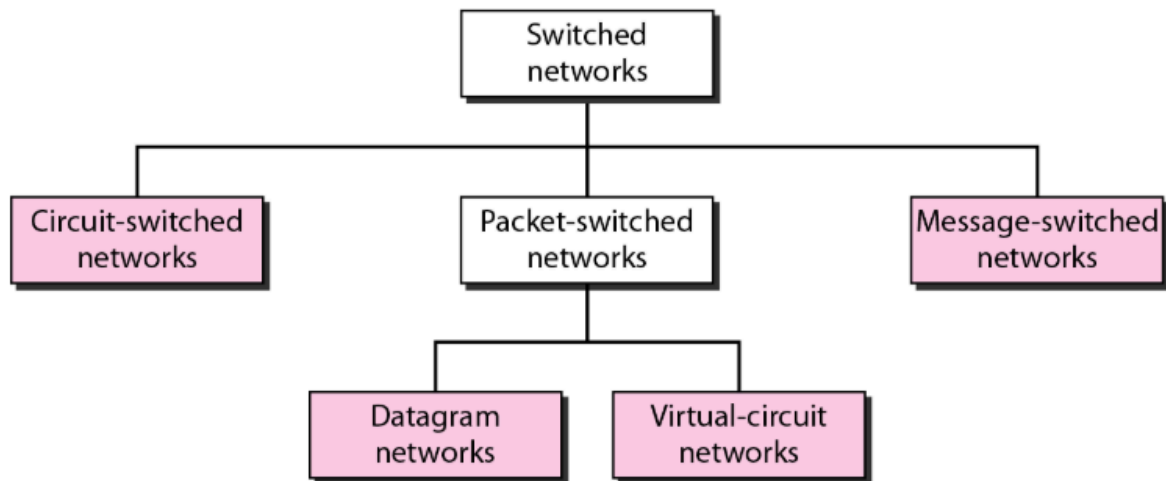


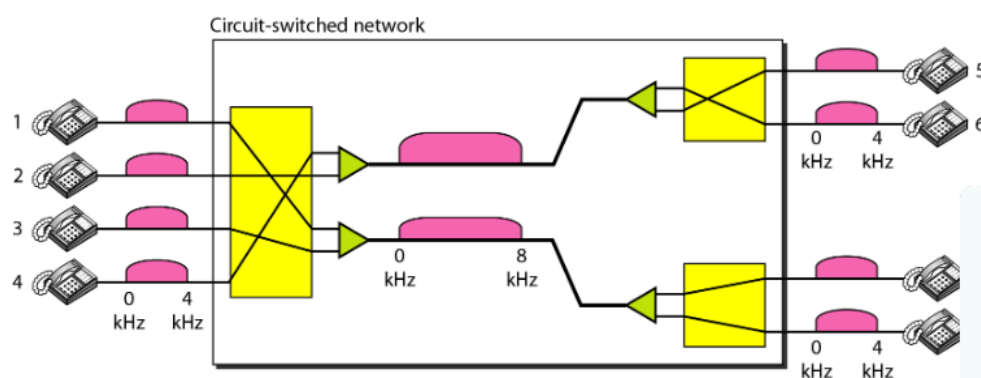
## Day 12 Of Communication System Class

### 2. Switching



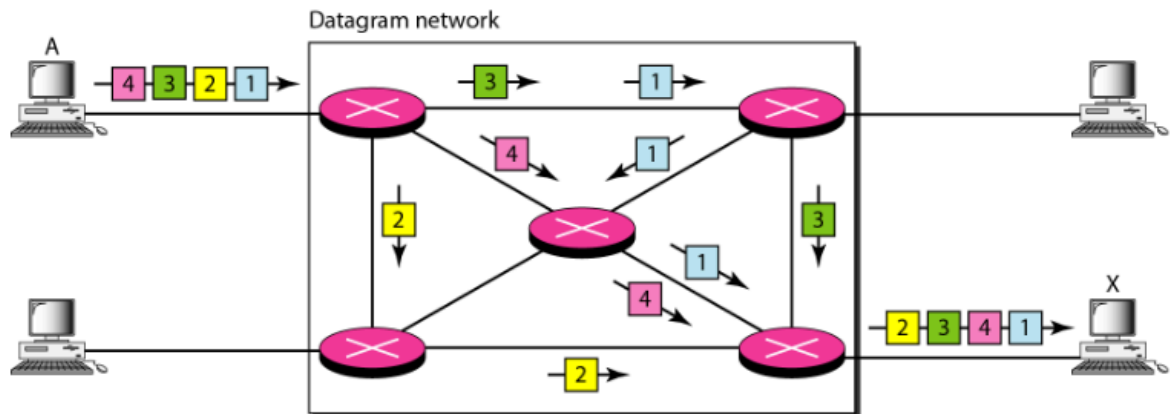
#### 2.1 Circuit Switching

**In circuit switching, the resources need to be reserved during the setup phase; the resources remain dedicated for the entire duration of data transfer until the teardown phase.**



## 2.2 Packet Switching

**In a packet-switched network, there is no resource reservation; resources are allocated on demand.**



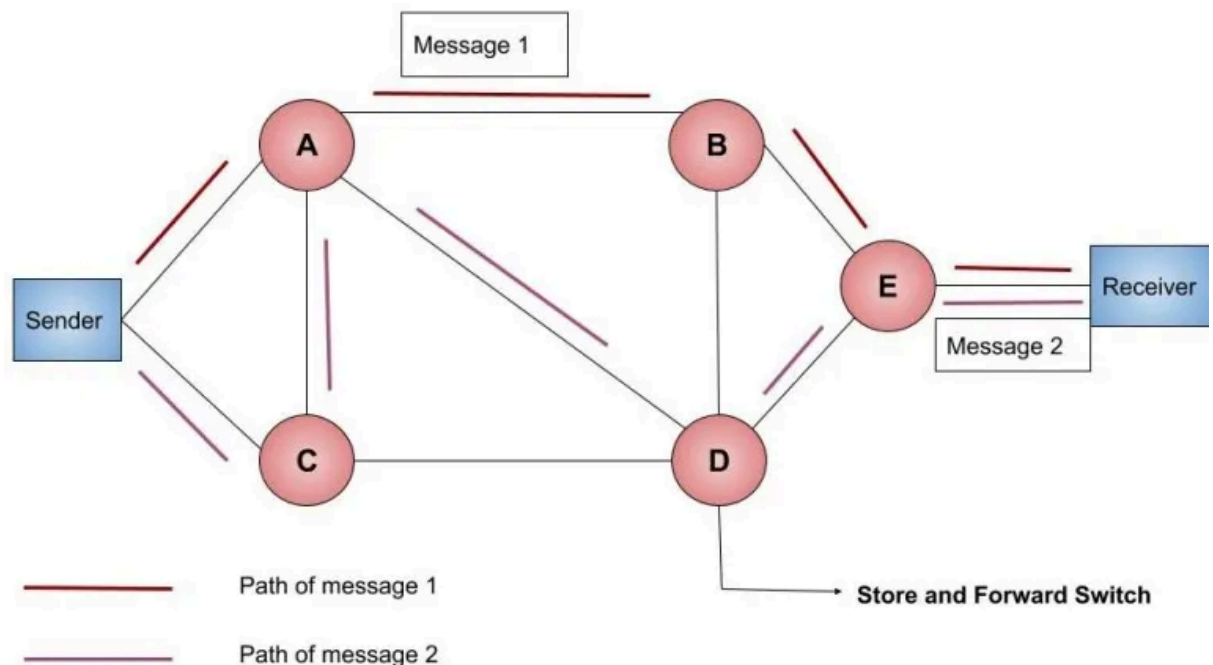
**Switching in the Internet is done by using the datagram approach to packet switching at the network layer.**

1. Compare packet switching and message switching.

[ Most Probable Question ]

| Circuit Switching                            | Packet Switching                        |
|----------------------------------------------|-----------------------------------------|
| Physical path between source and destination | No physical path                        |
| All packets use same path                    | Packets travel independently            |
| Reserve the entire bandwidth in advance      | Does not reserve                        |
| Bandwidth Wastage                            | No Bandwidth wastage                    |
| No store and forward transmission            | Supports store and forward transmission |

### 2.3 Message Switching



**Message Switching**

### 3. Multiple Access

***Channelization*** is a multiple-access method in which the available bandwidth of a link is shared in time, frequency, or through code, between different stations. In this section, we discuss three channelization protocols.

**Topics discussed in this section:**

**Frequency-Division Multiple Access (FDMA)**

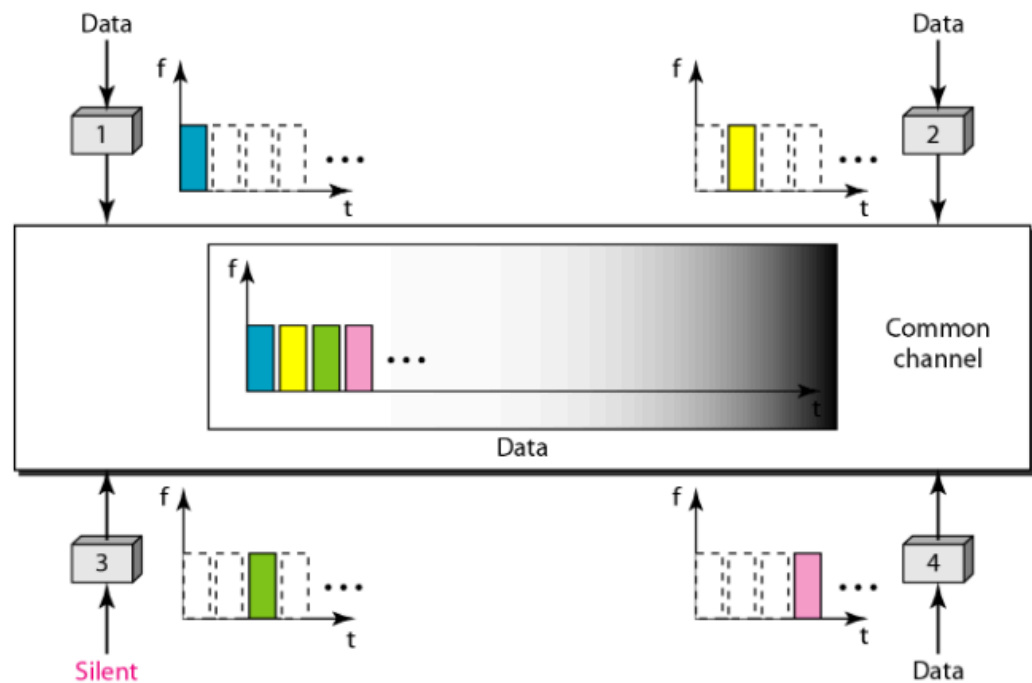
**Time-Division Multiple Access (TDMA)**

**Code-Division Multiple Access (CDMA)**

### 3.1 TDMA

In TDMA, the bandwidth is just one channel that is timeshared between different stations.

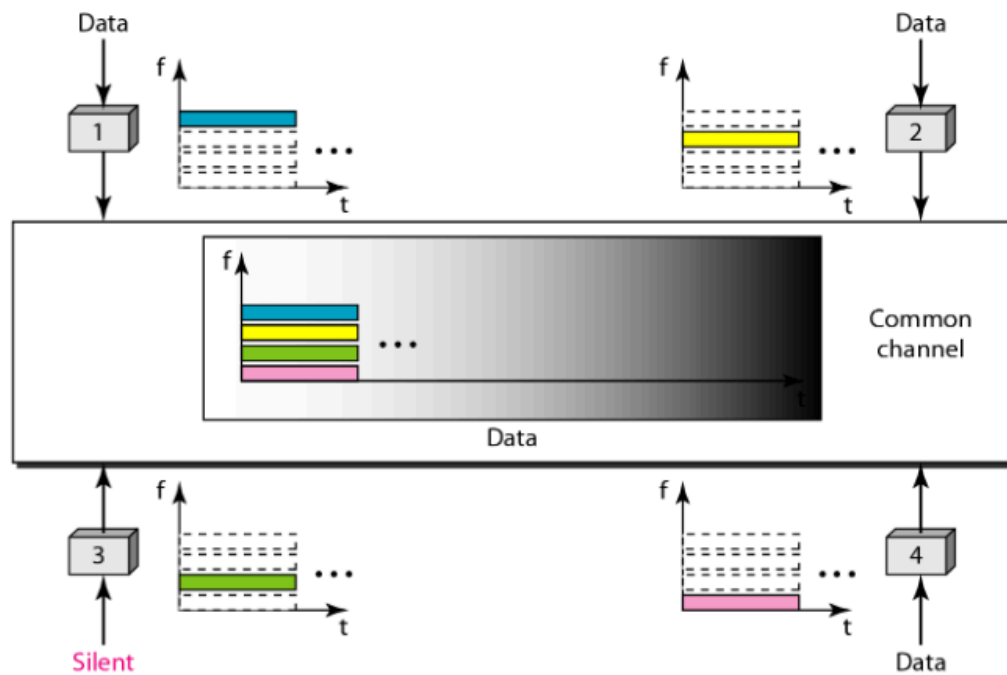
**Figure 12.22** *Time-division multiple access (TDMA)*



### 3.2 FDMA

In FDMA, the available bandwidth of the common channel is divided into bands that are separated by guard bands.

**Figure 12.21** *Frequency-division multiple access (FDMA)*



### 3.3 SDMA

### 3.4 CDMA

In CDMA, one channel carries all transmissions simultaneously.

**Figure 12.23** *Simple idea of communication with code*

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